

Linear Regression Model for Council District IV

Elliot Dean - Program Analyst IV
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

1 April 2026

Contents

Model I	1
Model II	5
Model III	9
Model IV	17

Model I

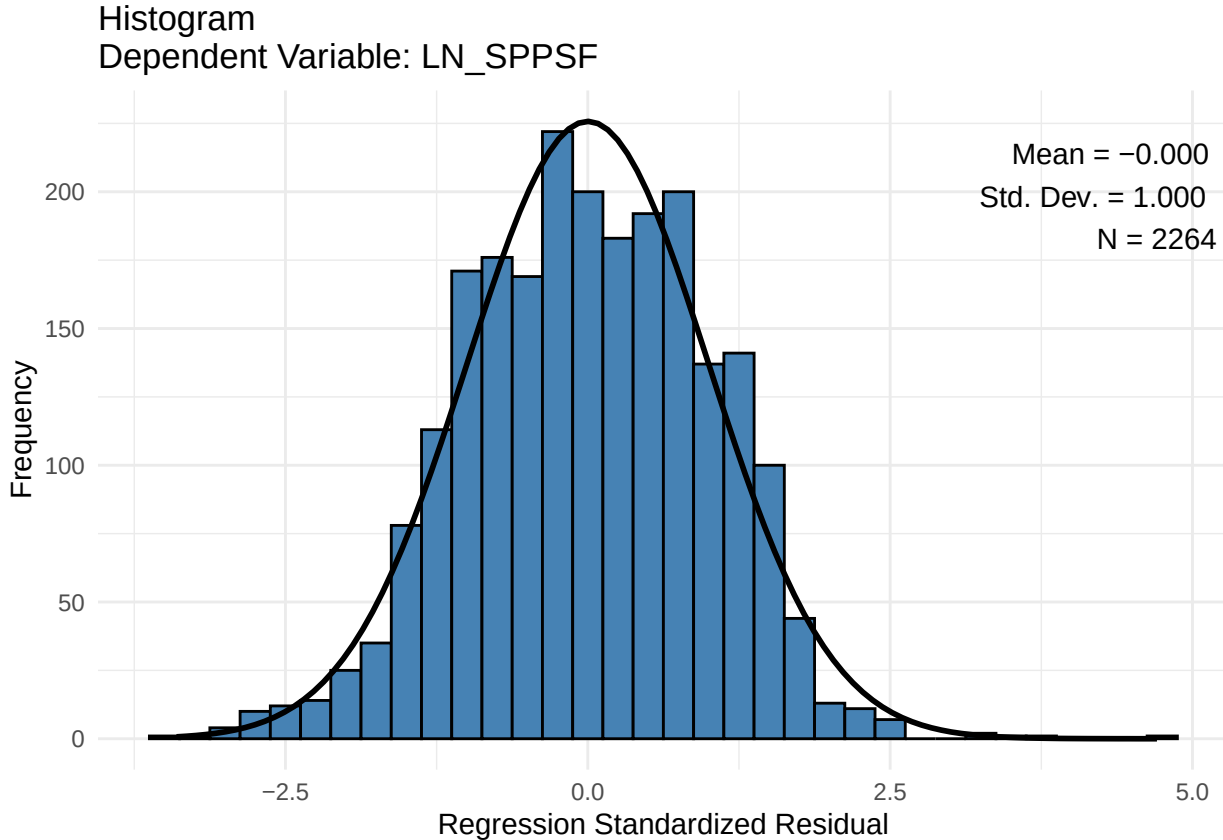


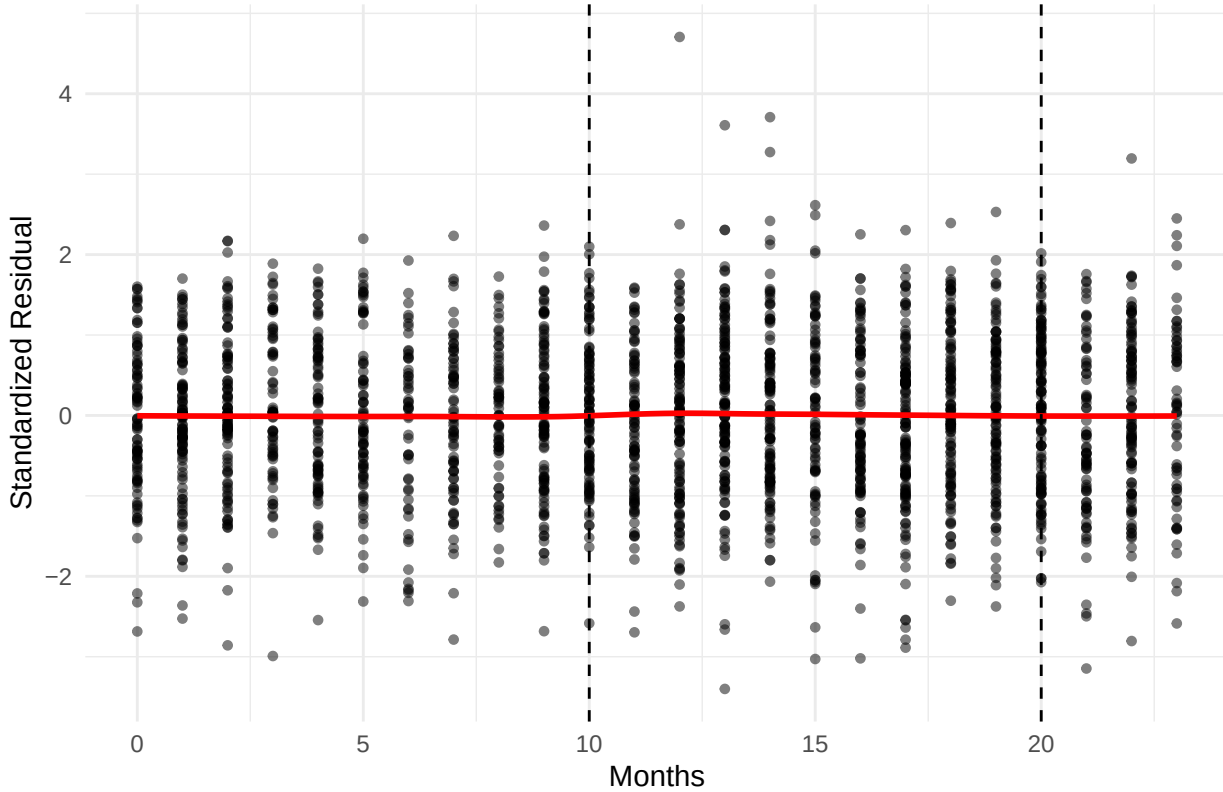
Table 1: District 4 LN_SPPSF model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2264	2245	19	0.5202	0.1163	0.1092	0.5181	0.421	16.4093	0	3486.963	3601.461

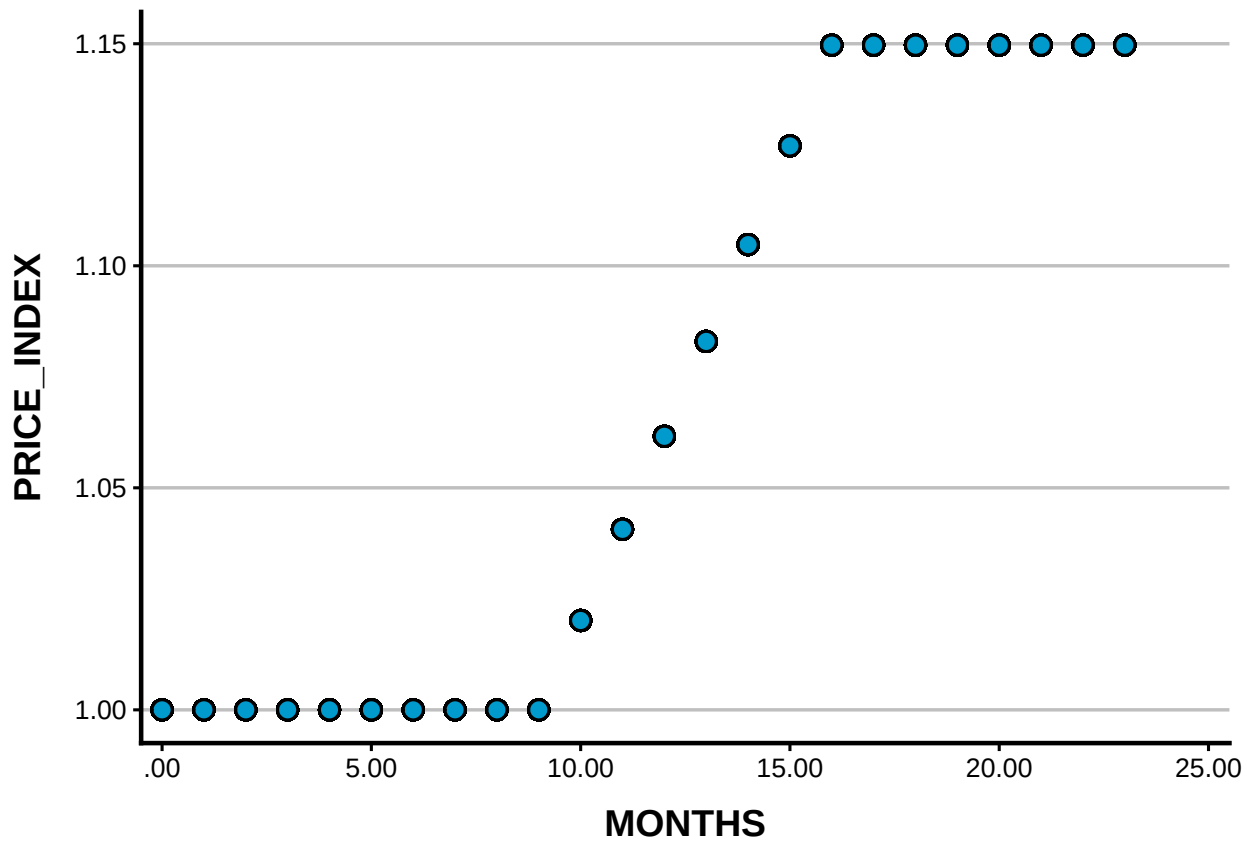
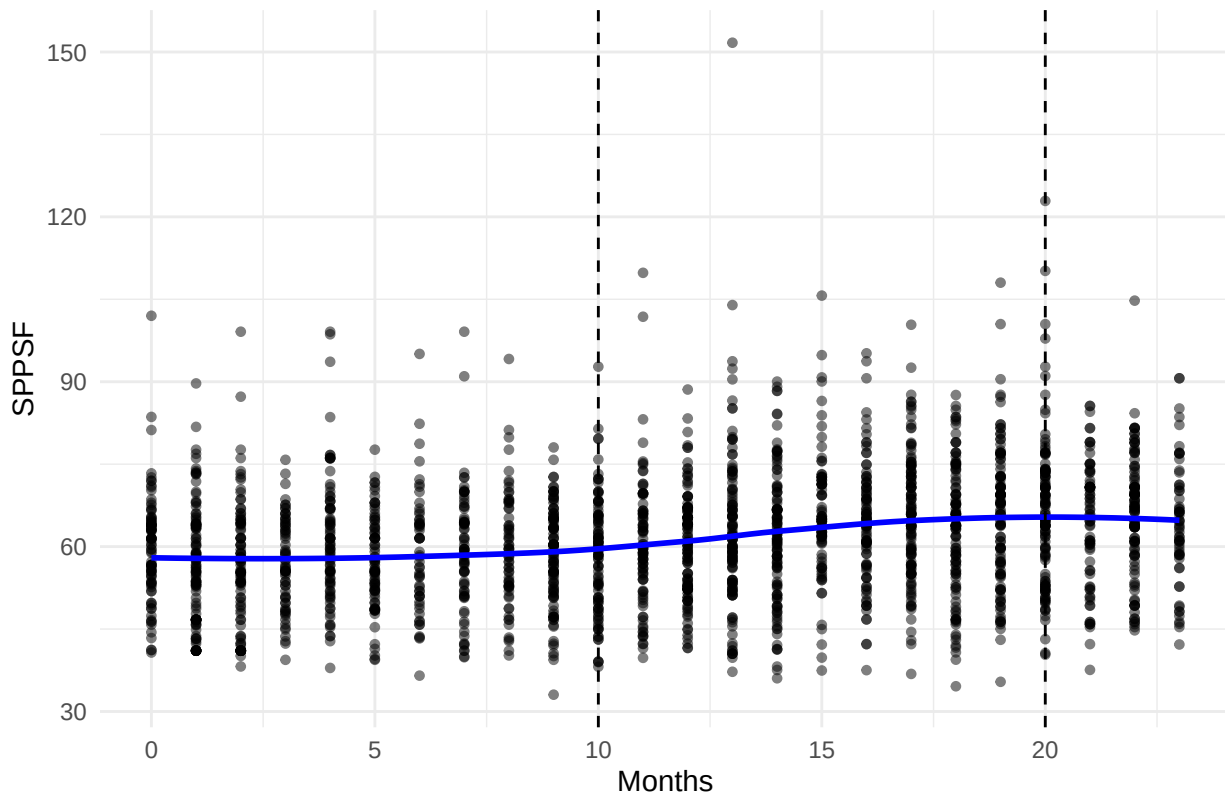
Table 2: District 4 LN_SPPSF model coefficients (95% CI)

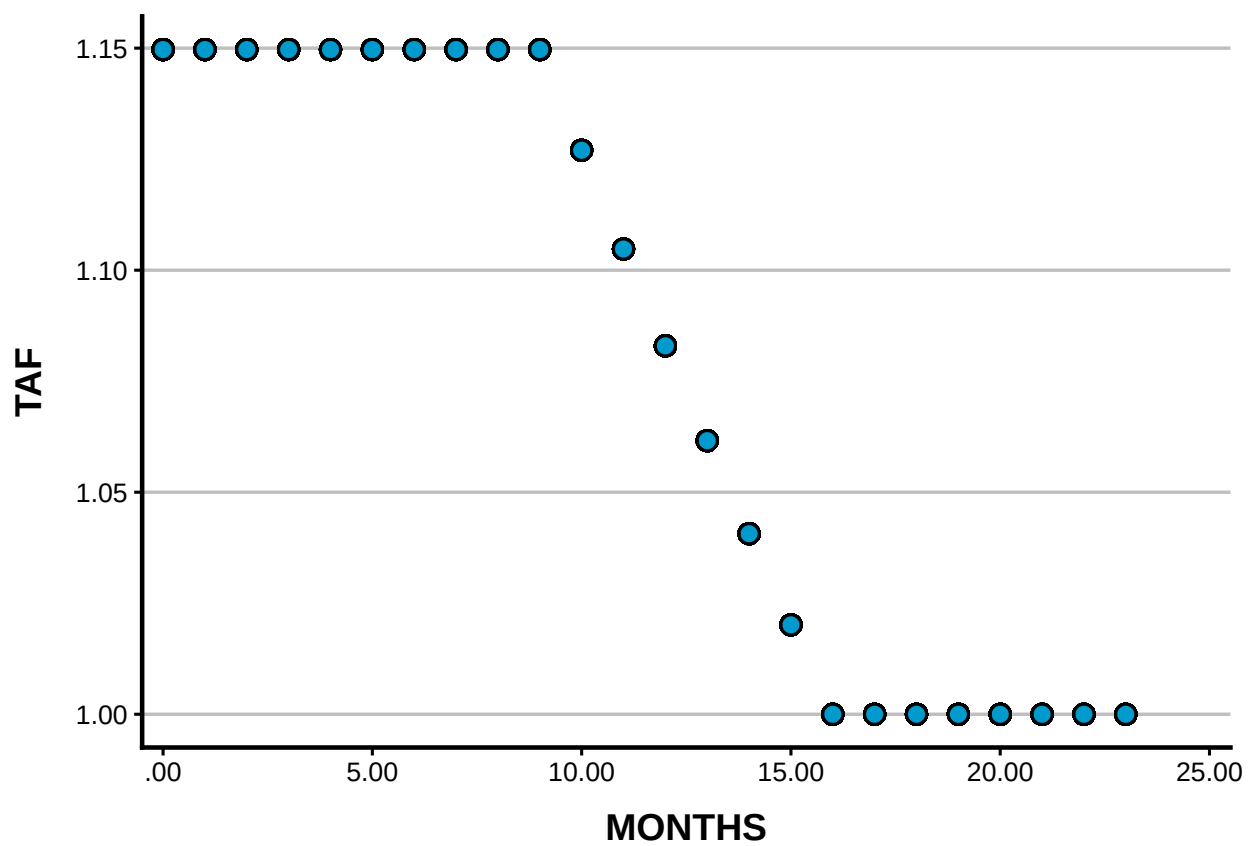
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	3.9389	98.3776	0.0400	0.0000	3.8604	4.0174
single_family1story	-0.0724	-2.5603	0.0283	0.0105	-0.1278	-0.0169
smallest	0.2781	5.8247	0.0477	0.0000	0.1845	0.3717
small	0.1358	3.7403	0.0363	0.0002	0.0646	0.2070
large	-0.0525	-1.6917	0.0310	0.0908	-0.1134	0.0084
largest	-0.1106	-2.3499	0.0471	0.0189	-0.2029	-0.0183
good	0.4351	4.5565	0.0955	0.0000	0.2478	0.6223
fair	-0.1927	-7.0605	0.0273	0.0000	-0.2463	-0.1392
poor	-0.3301	-4.8646	0.0678	0.0000	-0.4631	-0.1970
very_poor_unsound	-0.3930	-3.3397	0.1177	0.0009	-0.6237	-0.1622
qabove_avg	0.3497	2.5468	0.1373	0.0109	0.0804	0.6189
qbelow_avg	-0.0709	-2.2088	0.0321	0.0273	-0.1338	-0.0080
yb_1946_to_1960	0.1040	3.7742	0.0276	0.0002	0.0500	0.1581
yb_post_1961	-0.2540	-3.9013	0.0651	0.0001	-0.3817	-0.1263
ln_garagespaces	0.1192	3.7831	0.0315	0.0002	0.0574	0.1810
fireplace	0.1504	5.9247	0.0254	0.0000	0.1006	0.2002
hot_water	0.0997	2.4563	0.0406	0.0141	0.0201	0.1794
central_air	0.1285	4.1907	0.0307	0.0000	0.0684	0.1887
months_10to16	0.0201	5.4934	0.0037	0.0000	0.0129	0.0273

Residuals for LN_SPPSF Model



Months vs Predicted SPPSF – LN_SPPSF Model





Model II

Histogram
Dependent Variable: LN_PRICE (No ECFs)

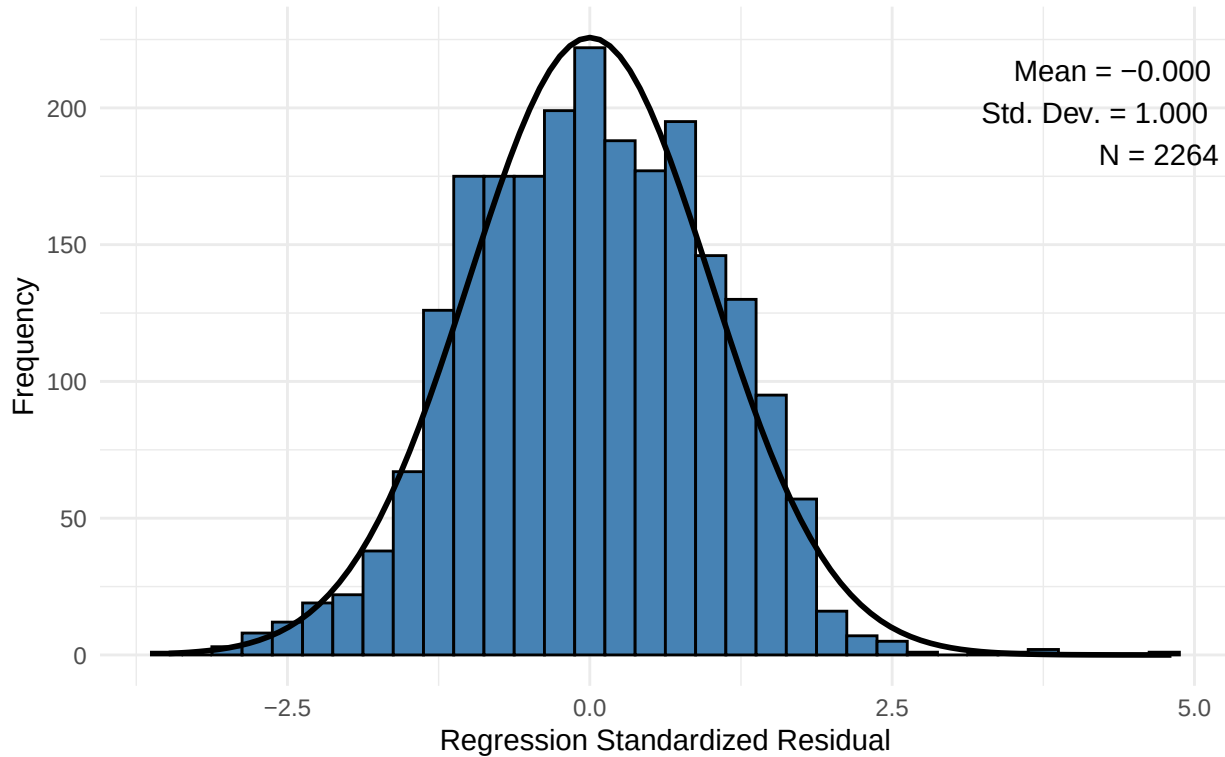


Table 3: District 4 LN_PRICE (No ECFs) model fit statistics

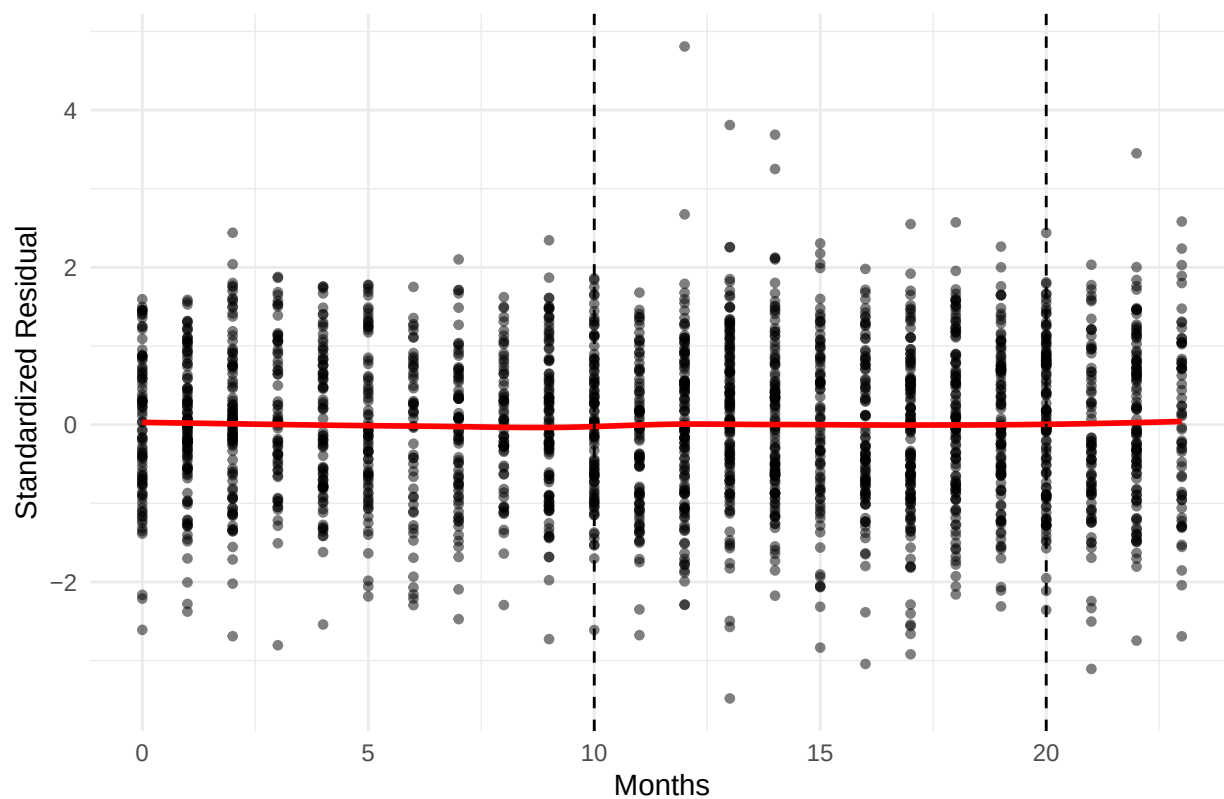
n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2264	2245	19	0.517	0.2584	0.2524	0.5149	0.4187	43.4496	0	3459.083	3573.581

Table 4: District 4 LN_PRICE (No ECFs) model coefficients (95% CI)

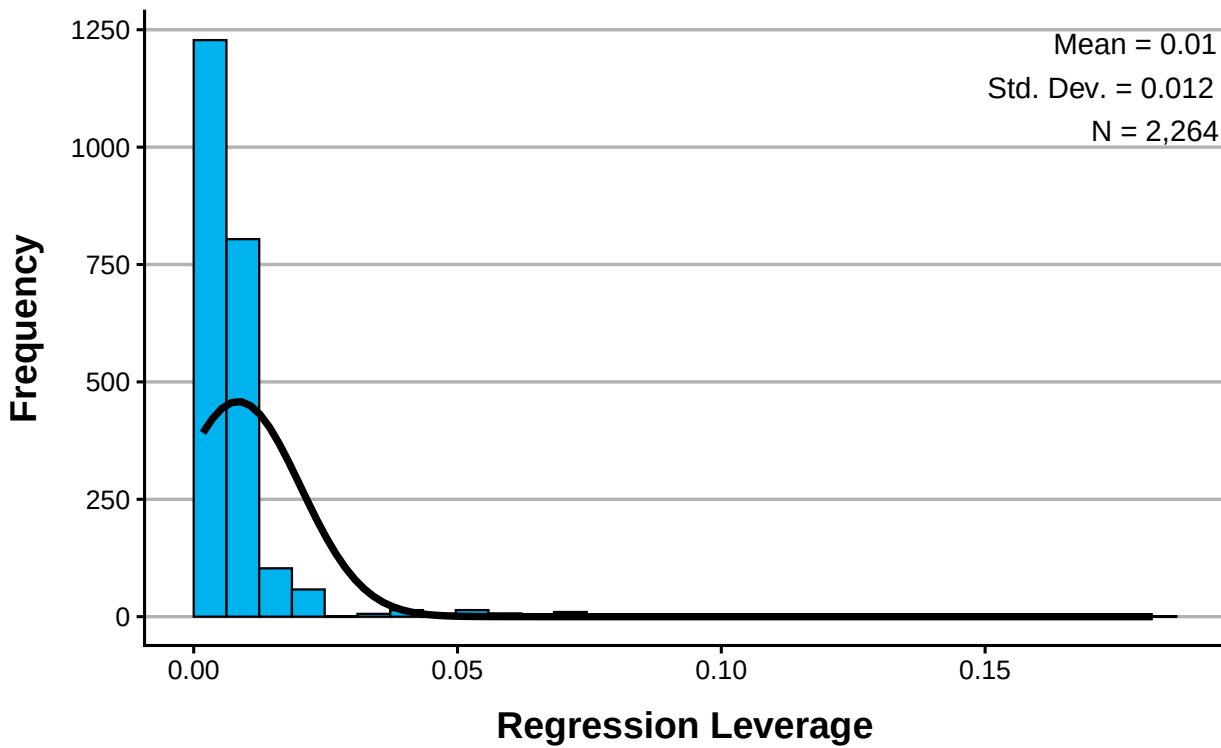
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.8515	305.3084	0.0355	0.0000	10.7818	10.9212
ln_landratio	0.3380	5.3758	0.0629	0.0000	0.2147	0.4612
ln_base_bldgsize_ratio	0.8502	11.0133	0.0772	0.0000	0.6988	1.0016
smallest	0.1853	3.2606	0.0568	0.0011	0.0739	0.2968
small	0.0961	2.4823	0.0387	0.0131	0.0202	0.1720
good	0.4419	4.5231	0.0977	0.0000	0.2503	0.6335
fair	-0.1853	-6.8317	0.0271	0.0000	-0.2385	-0.1321
poor	-0.3221	-4.7774	0.0674	0.0000	-0.4543	-0.1899
very_poor_unsound	-0.3804	-3.2524	0.1170	0.0012	-0.6098	-0.1510
qabove_avg	0.3244	2.3700	0.1369	0.0179	0.0560	0.5928
qbelow_avg	-0.0702	-2.1911	0.0320	0.0286	-0.1331	-0.0074
yb_1946_to_1960	0.0857	3.0998	0.0276	0.0020	0.0315	0.1399
yb_post_1961	-0.2329	-3.6159	0.0644	0.0003	-0.3592	-0.1066
slab	-0.3128	-1.7220	0.1817	0.0852	-0.6691	0.0434
ln_garagespaces	0.0991	3.1276	0.0317	0.0018	0.0370	0.1613
fireplace	0.1419	5.5929	0.0254	0.0000	0.0921	0.1916
hot_water	0.0955	2.3848	0.0400	0.0172	0.0170	0.1740

term	estimate	statistic	std_error	p_value	conf_low	conf_high
central_air	0.1190	3.9046	0.0305	0.0001	0.0592	0.1788
months_10to16	0.0192	5.3571	0.0036	0.0000	0.0121	0.0262

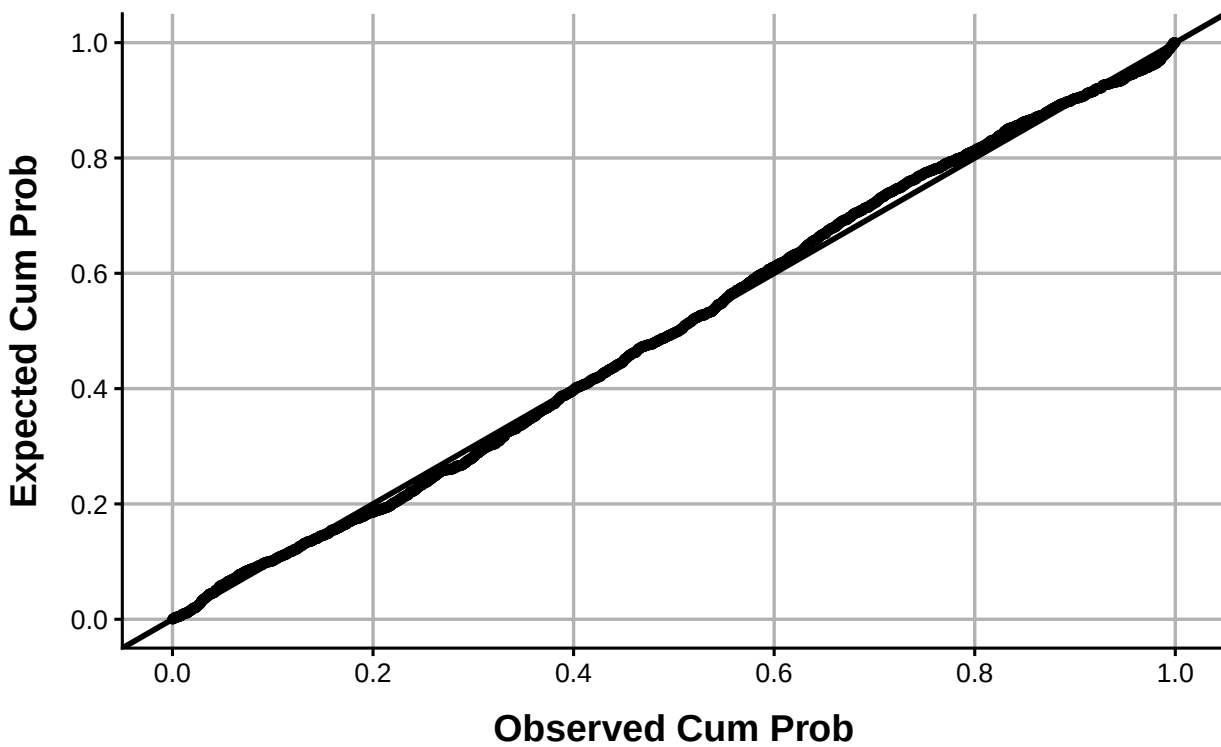
Residuals for LN_PRICE (No ECFs) Model

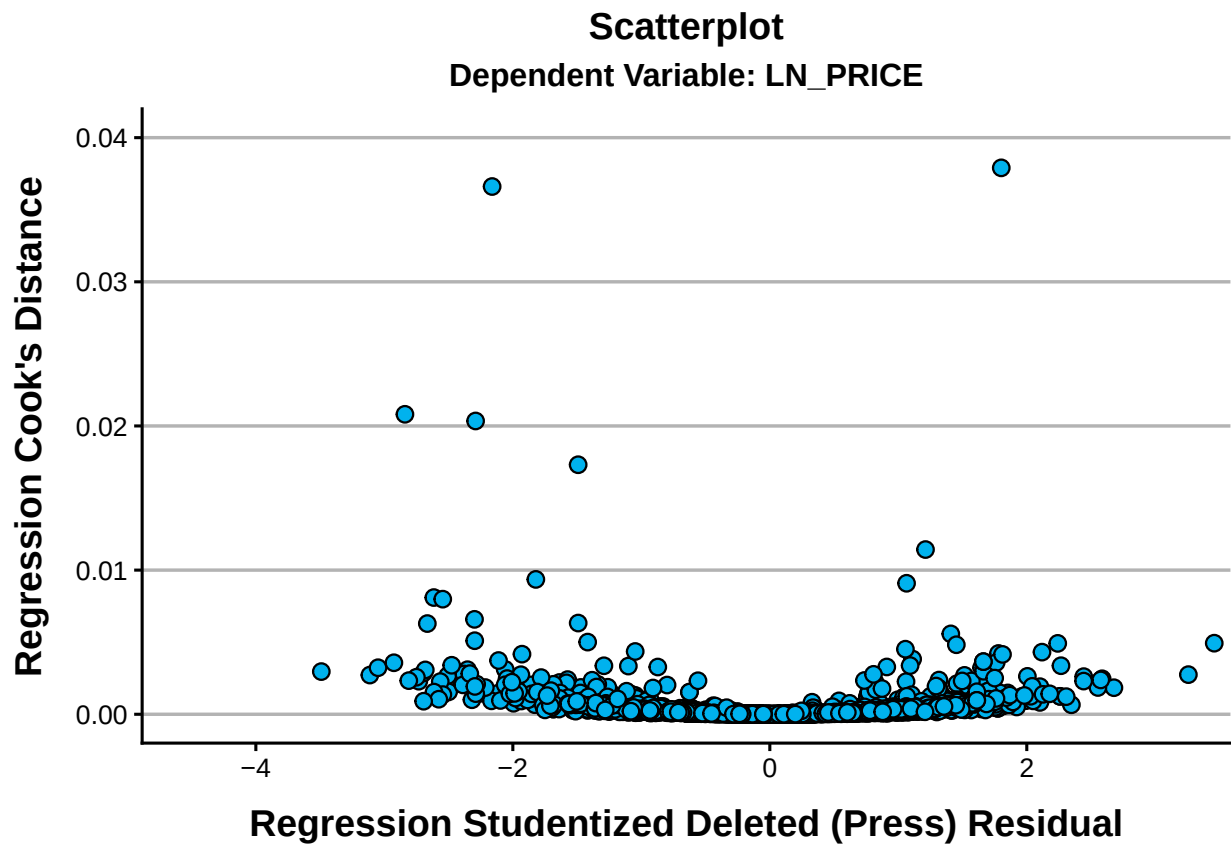
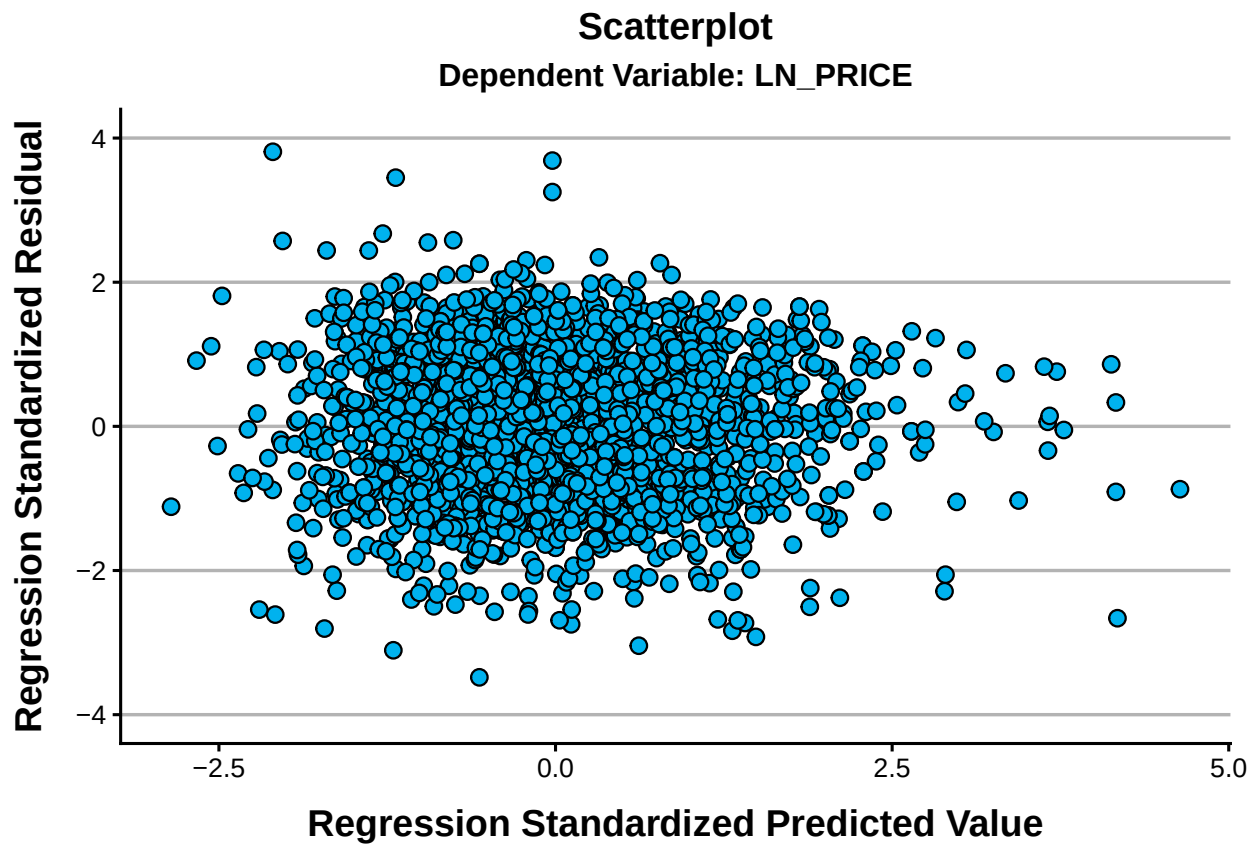


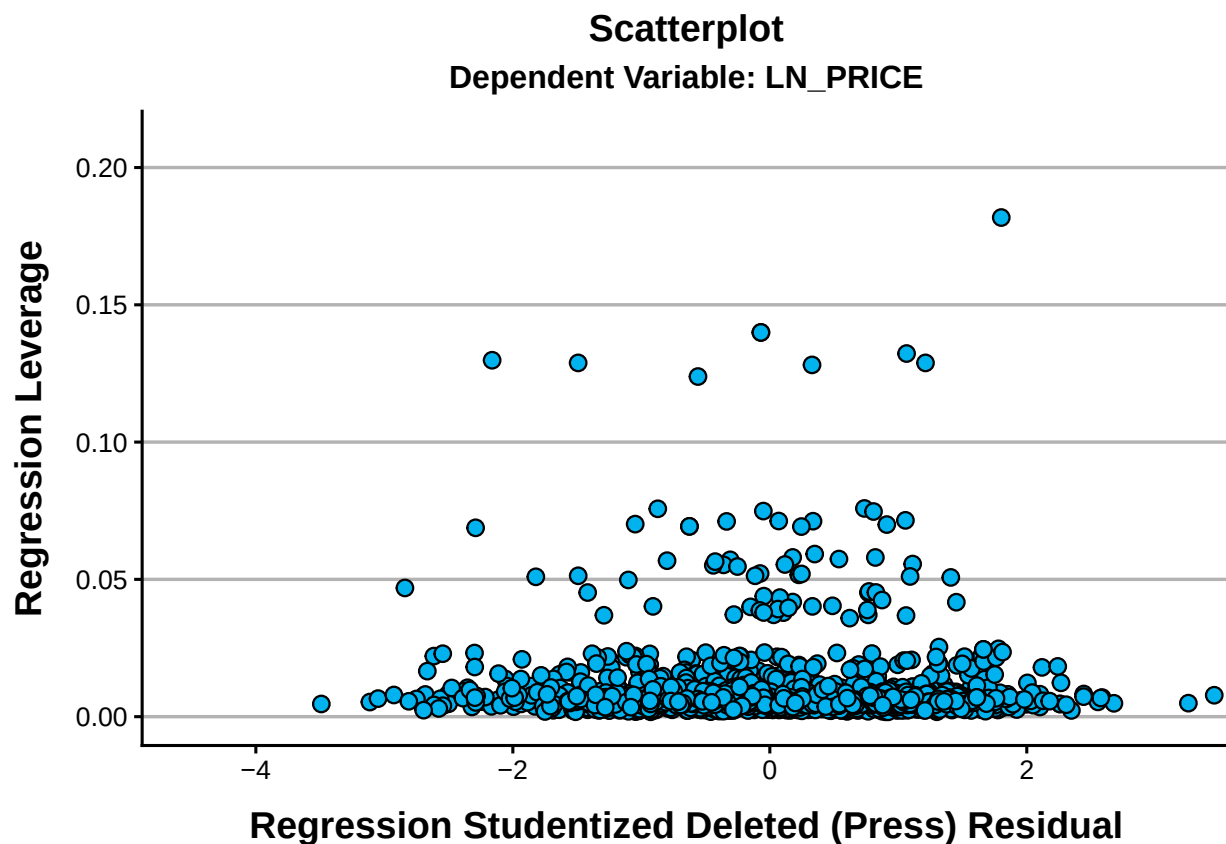
Histogram
Dependent Variable: LN_PRICE



Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE







Model III

Initial Input

Table 5: LN_PRICE model (base) assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2264	0.9956138	1.112098	0.902648	1.23204	40.95043	-0.6702013

Table 6: District 4 LN_PRICE model fit statistics

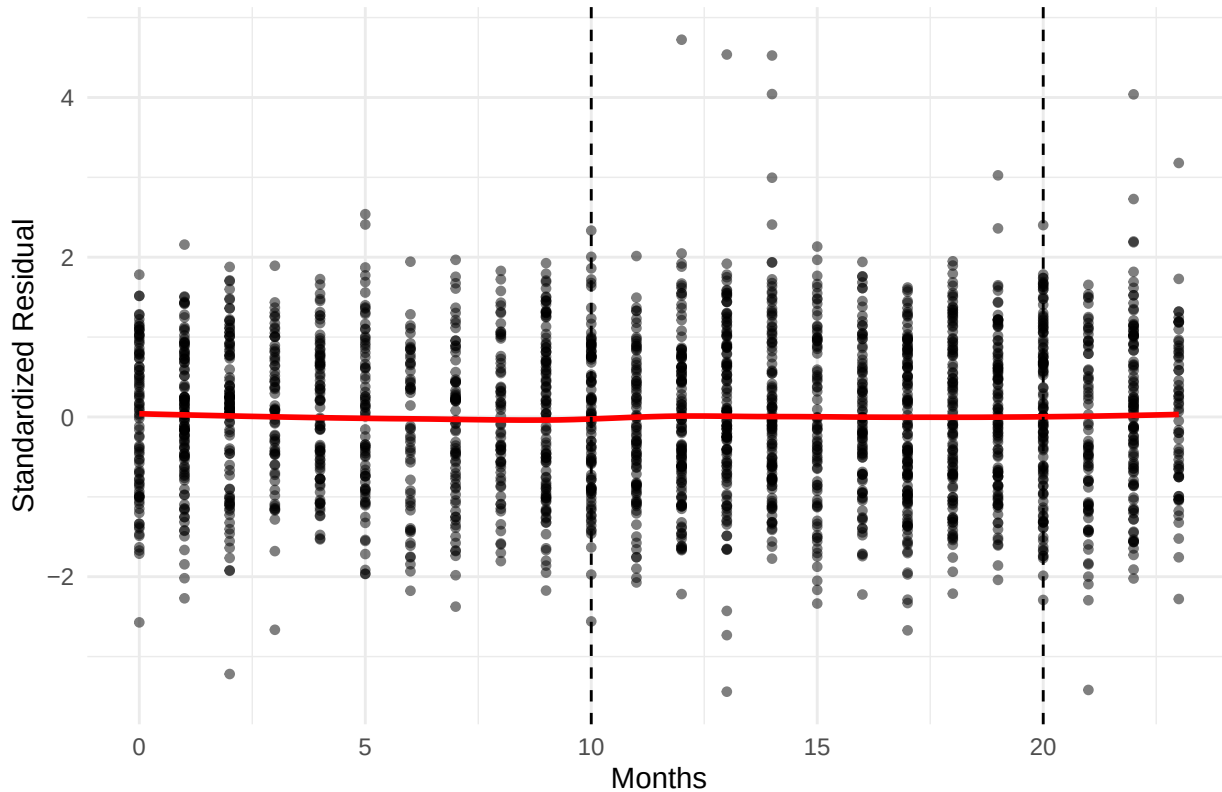
n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2264	2233	31	0.4676	0.3967	0.3886	0.4644	0.3823	48.9421	0	3015.723	3198.92

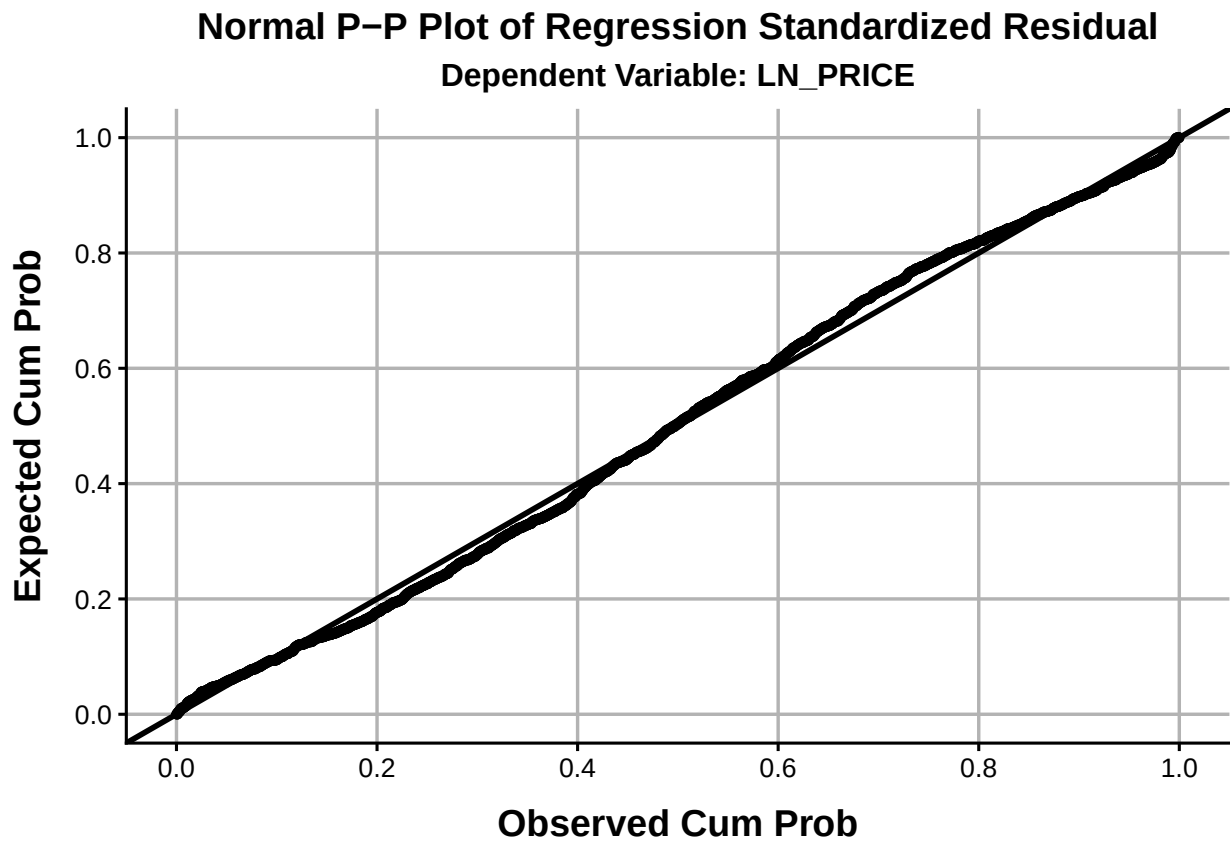
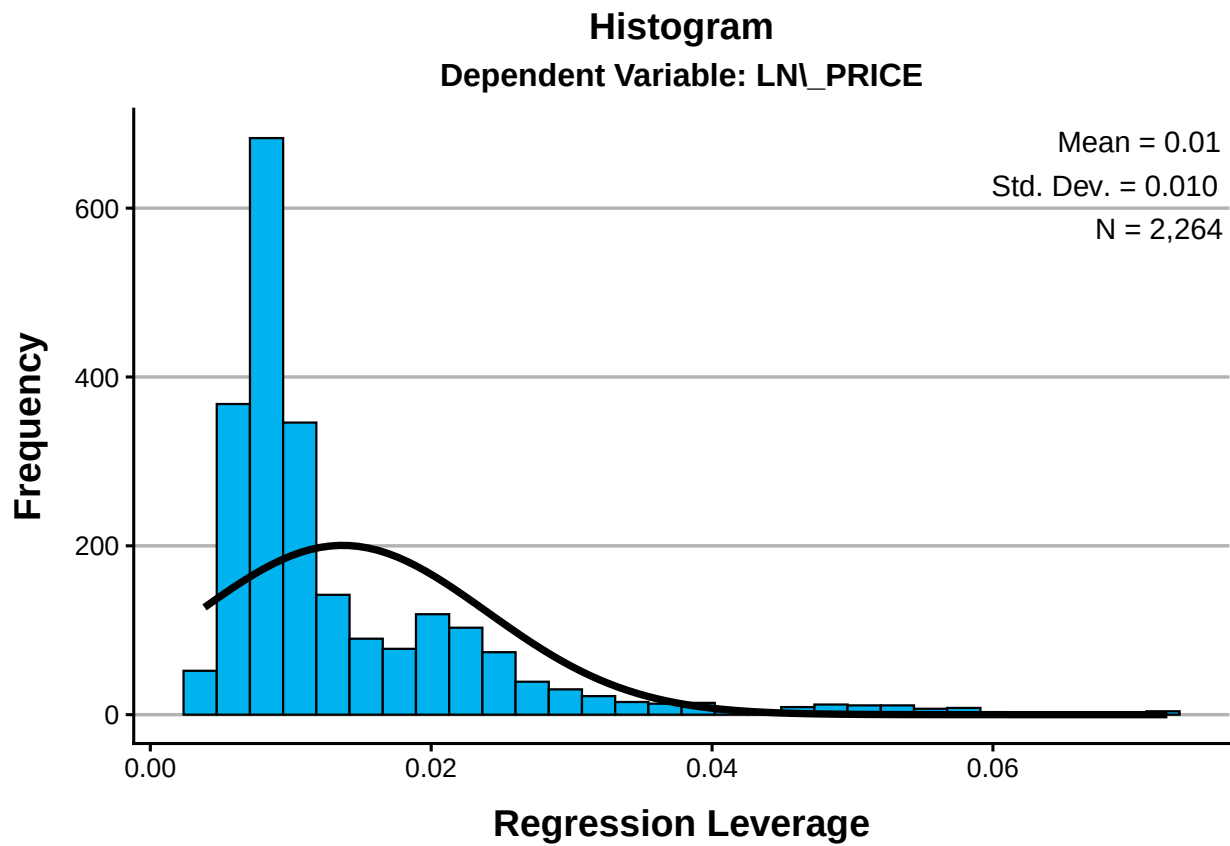
Table 7: District 4 LN_PRICE model coefficients (95% CI)

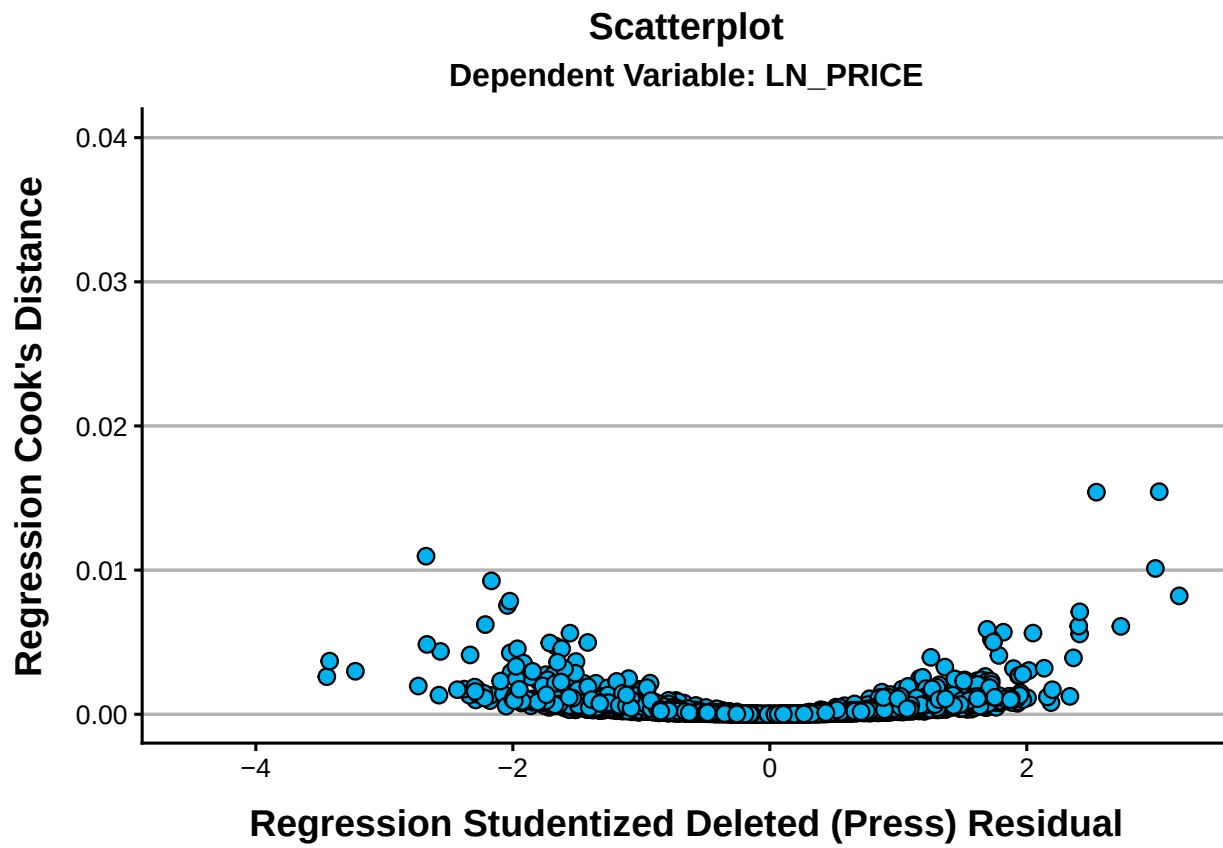
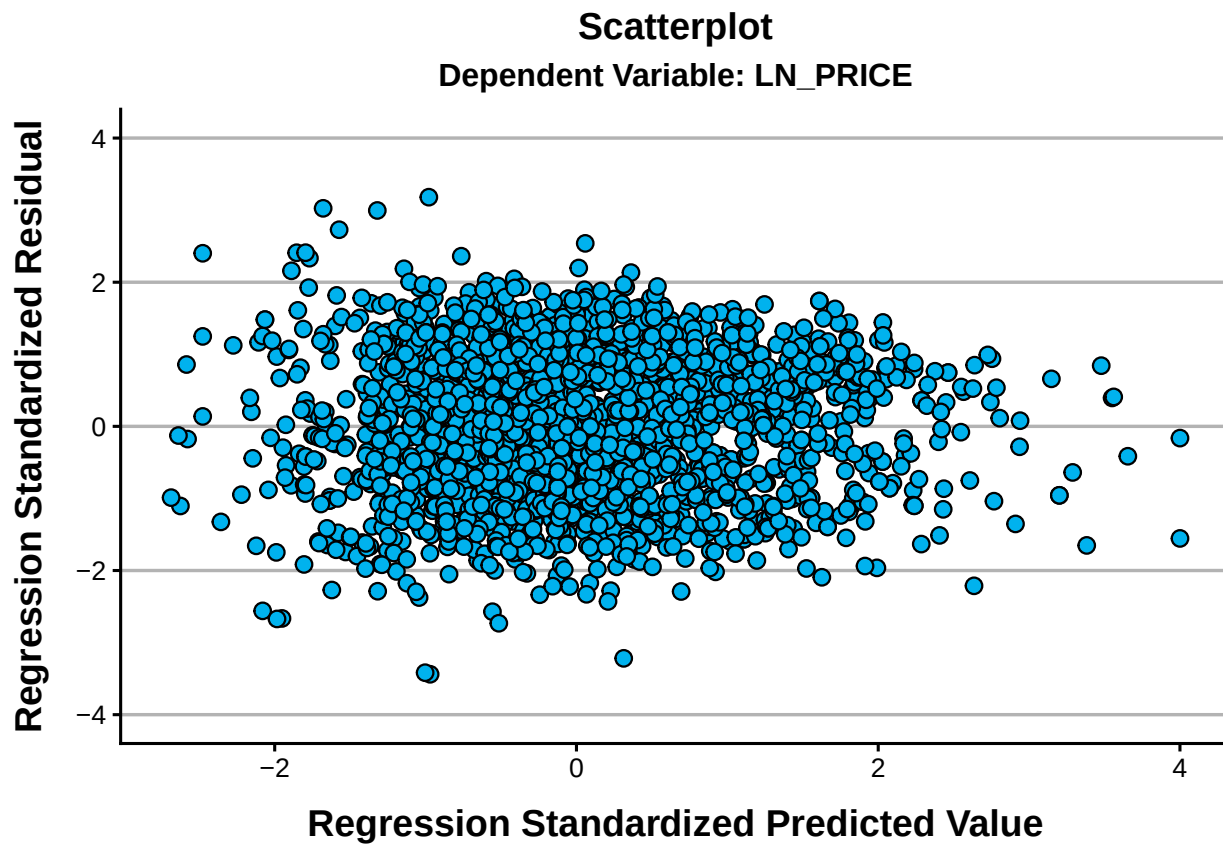
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.7783	287.0813	0.0375	0.0000	10.7047	10.8520
ln_landratio	0.1087	1.8222	0.0596	0.0686	-0.0083	0.2257
ln_base_bldgsize_ratio	0.6150	10.5342	0.0584	0.0000	0.5005	0.7295
arterial	0.0678	1.9419	0.0349	0.0523	-0.0007	0.1363
single_family1story	-0.0676	-2.4566	0.0275	0.0141	-0.1216	-0.0136
half_duplex1story	-0.1949	-2.6866	0.0725	0.0073	-0.3371	-0.0526

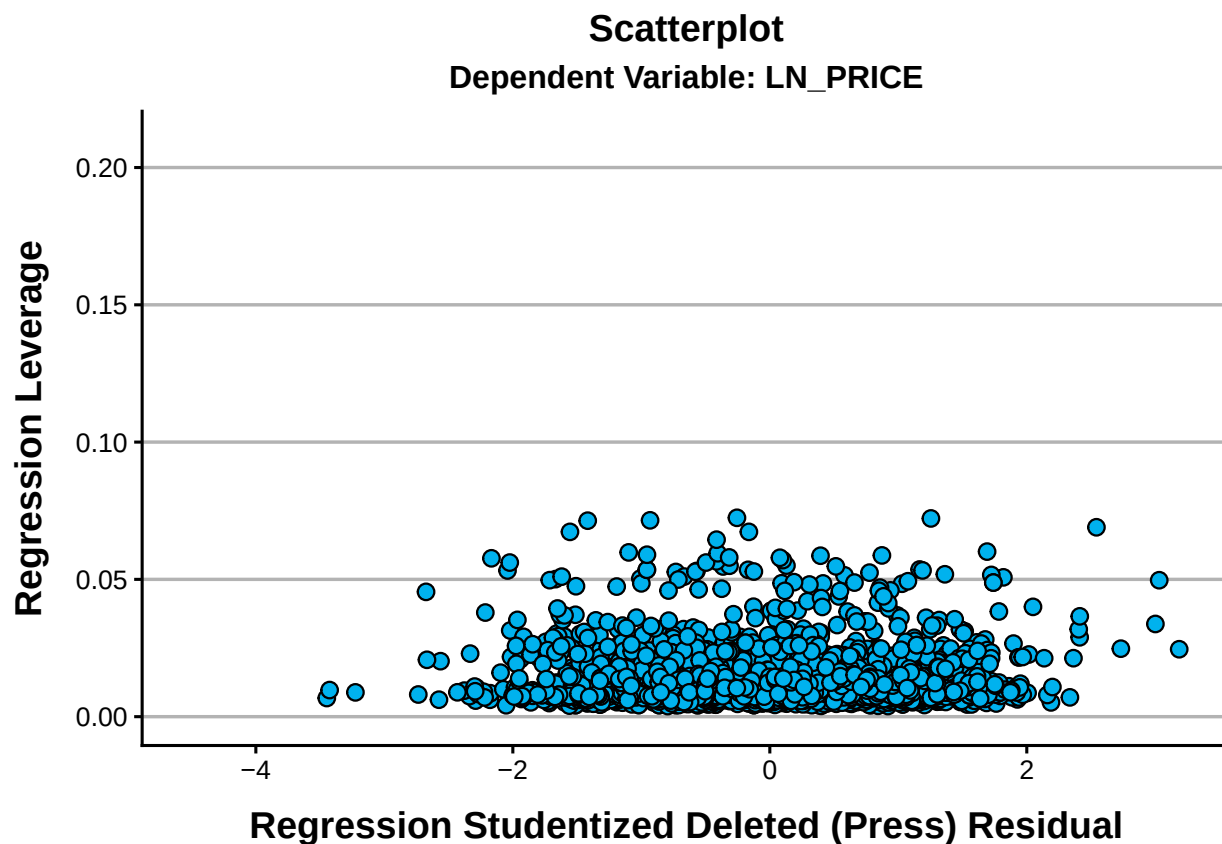
term	estimate	statistic	std_error	p_value	conf_low	conf_high
two_plus_baths	0.1384	1.7378	0.0796	0.0824	-0.0178	0.2946
good	0.2774	3.2082	0.0865	0.0014	0.1078	0.4469
fair	-0.1781	-7.2064	0.0247	0.0000	-0.2266	-0.1297
poor	-0.2656	-4.3341	0.0613	0.0000	-0.3858	-0.1454
very_poor_unsound	-0.2226	-2.0915	0.1064	0.0366	-0.4313	-0.0139
yb_1946_to_1960	0.0584	2.2838	0.0256	0.0225	0.0082	0.1085
ln_garagespaces	0.0704	2.4412	0.0289	0.0147	0.0139	0.1270
fireplace	0.0815	3.4798	0.0234	0.0005	0.0356	0.1274
central_air	0.0850	3.1013	0.0274	0.0020	0.0312	0.1387
ecf_4R402	-0.4063	-5.3807	0.0755	0.0000	-0.5544	-0.2582
ecf_4R404	0.1271	2.3398	0.0543	0.0194	0.0206	0.2336
ecf_4R405	0.2391	5.9487	0.0402	0.0000	0.1603	0.3179
ecf_4R406	0.1855	4.7889	0.0387	0.0000	0.1096	0.2615
ecf_4R408	0.0858	1.9820	0.0433	0.0476	0.0009	0.1707
ecf_4R409	0.1587	2.4393	0.0651	0.0148	0.0311	0.2863
ecf_4R410	0.4615	12.5694	0.0367	0.0000	0.3895	0.5335
ecf_4R413	0.2969	6.9937	0.0425	0.0000	0.2137	0.3802
ecf_4R414	0.6385	10.7063	0.0596	0.0000	0.5216	0.7555
ecf_4R415	0.7986	11.5939	0.0689	0.0000	0.6635	0.9337
ecf_4R416	-0.2420	-2.3571	0.1027	0.0185	-0.4433	-0.0407
ecf_4R417	0.4437	10.4945	0.0423	0.0000	0.3608	0.5266
ecf_4R419	-0.1719	-2.5973	0.0662	0.0095	-0.3017	-0.0421
ecf_4R421	-0.1706	-2.1646	0.0788	0.0305	-0.3252	-0.0160
ecf_4R422	0.6487	11.0190	0.0589	0.0000	0.5332	0.7641
months_10to16	0.0225	6.7857	0.0033	0.0000	0.0160	0.0289

Residuals for LN_PRICE Model









Filtered Input

Table 8: District 4 LN_PRICE (Inliers) model fit statistics

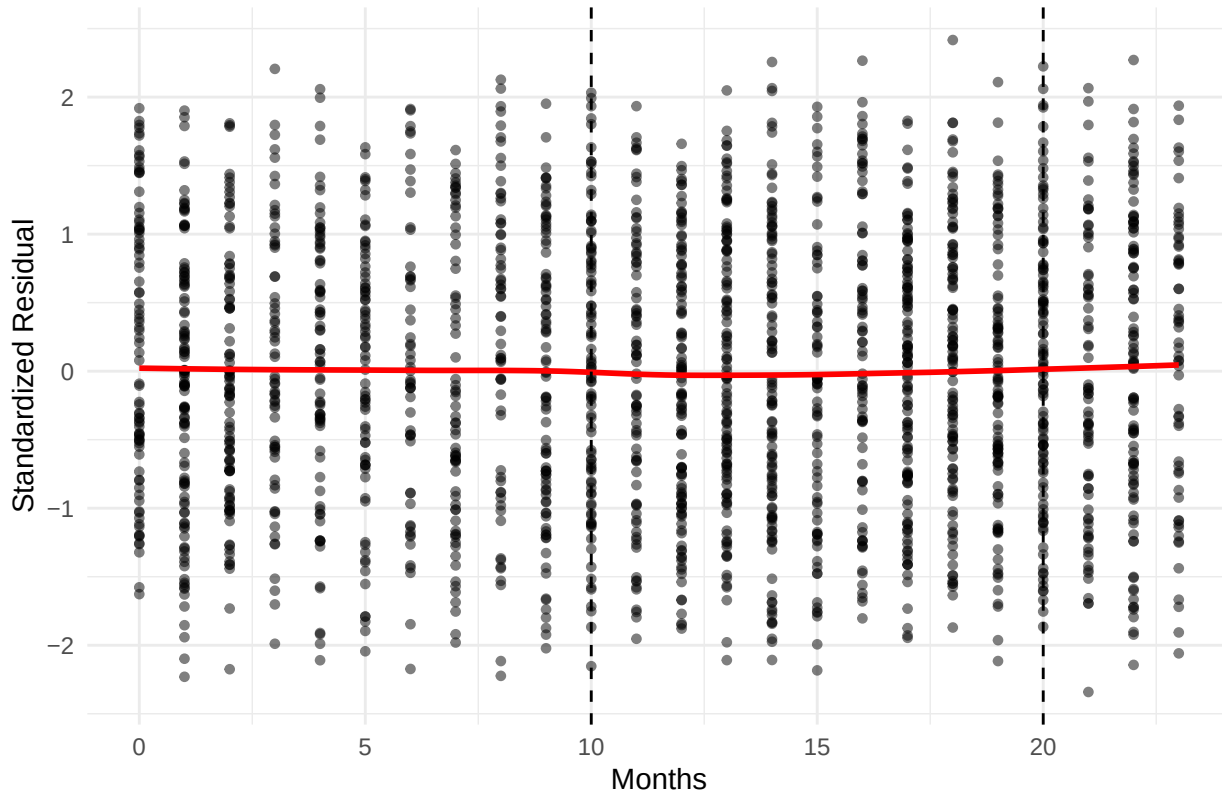
n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2130	2100	30	0.4144	0.454	0.4464	0.4115	0.347	60.2054	0	2323.839	2499.419

Table 9: District 4 LN_PRICE (Inliers) model coefficients (95% CI)

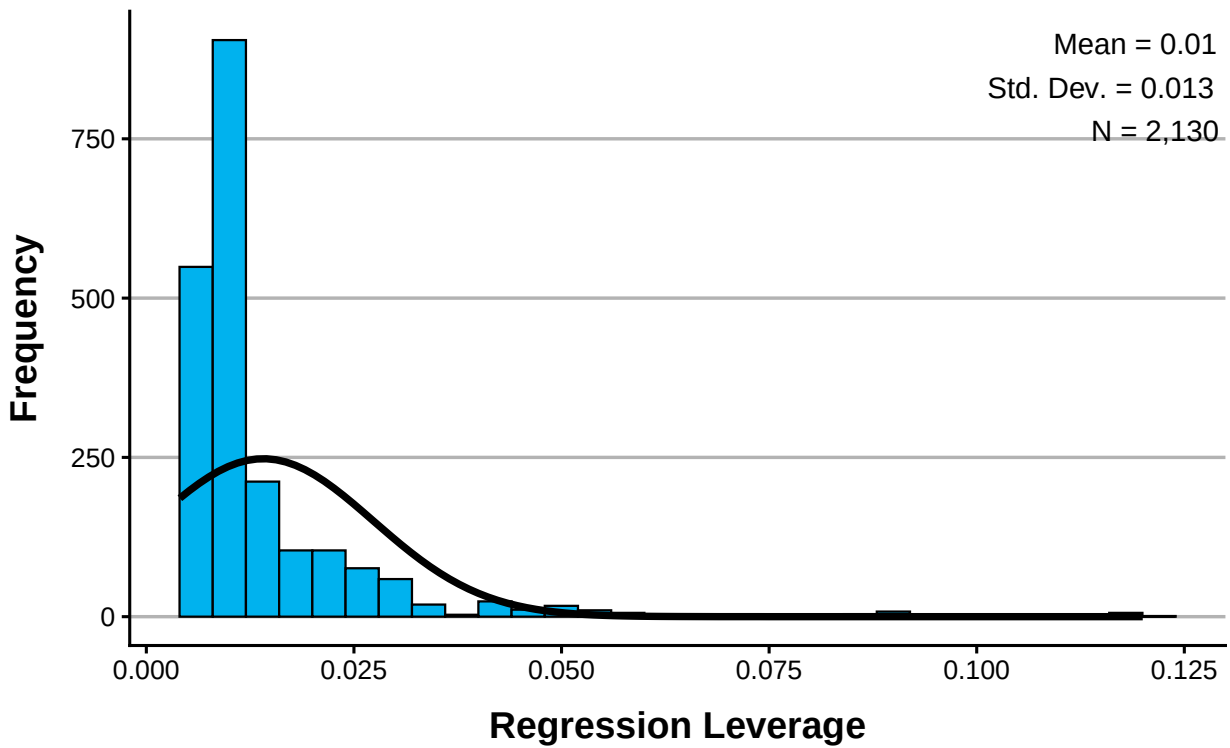
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.8138	310.9014	0.0348	0.0000	10.7456	10.8821
ln_landratio	0.1107	2.0022	0.0553	0.0454	0.0023	0.2192
ln_base_bldgsize_ratio	0.5475	9.0896	0.0602	0.0000	0.4294	0.6656
single_family1story	-0.0752	-3.0156	0.0249	0.0026	-0.1241	-0.0263
half_duplex1story	-0.1673	-2.4517	0.0682	0.0143	-0.3012	-0.0335
two_plus_baths	0.2438	2.7671	0.0881	0.0057	0.0710	0.4166
good	0.2884	3.3631	0.0858	0.0008	0.1202	0.4566
fair	-0.1636	-7.1864	0.0228	0.0000	-0.2082	-0.1190
poor	-0.2467	-3.7733	0.0654	0.0002	-0.3748	-0.1185
very_poor_unsound	-0.3278	-2.7074	0.1211	0.0068	-0.5652	-0.0904
qbelow_avg	-0.0608	-2.3152	0.0263	0.0207	-0.1124	-0.0093
yb_1946_to_1960	0.0491	2.1325	0.0230	0.0331	0.0039	0.0942
ln_garagespaces	0.0659	2.4602	0.0268	0.0140	0.0134	0.1185
fireplace	0.0947	4.4661	0.0212	0.0000	0.0531	0.1363
wall	-0.2829	-2.0028	0.1412	0.0453	-0.5599	-0.0059
central_air	0.0887	3.6020	0.0246	0.0003	0.0404	0.1369

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_4R402	-0.4778	-5.7898	0.0825	0.0000	-0.6397	-0.3160
ecf_4R404	0.1863	3.7397	0.0498	0.0002	0.0886	0.2841
ecf_4R405	0.2491	6.9702	0.0357	0.0000	0.1790	0.3192
ecf_4R406	0.1736	5.0443	0.0344	0.0000	0.1061	0.2410
ecf_4R408	0.0966	2.5137	0.0384	0.0120	0.0212	0.1719
ecf_4R409	0.1567	2.6161	0.0599	0.0090	0.0392	0.2742
ecf_4R410	0.4527	13.7580	0.0329	0.0000	0.3882	0.5173
ecf_4R413	0.2860	7.4969	0.0382	0.0000	0.2112	0.3609
ecf_4R414	0.6962	12.8346	0.0542	0.0000	0.5899	0.8026
ecf_4R415	0.7635	12.2013	0.0626	0.0000	0.6408	0.8862
ecf_4R417	0.4543	12.0652	0.0376	0.0000	0.3804	0.5281
ecf_4R419	-0.2215	-3.4522	0.0642	0.0006	-0.3474	-0.0957
ecf_4R422	0.7113	13.2522	0.0537	0.0000	0.6060	0.8165
months_10to16	0.0214	7.1397	0.0030	0.0000	0.0156	0.0273

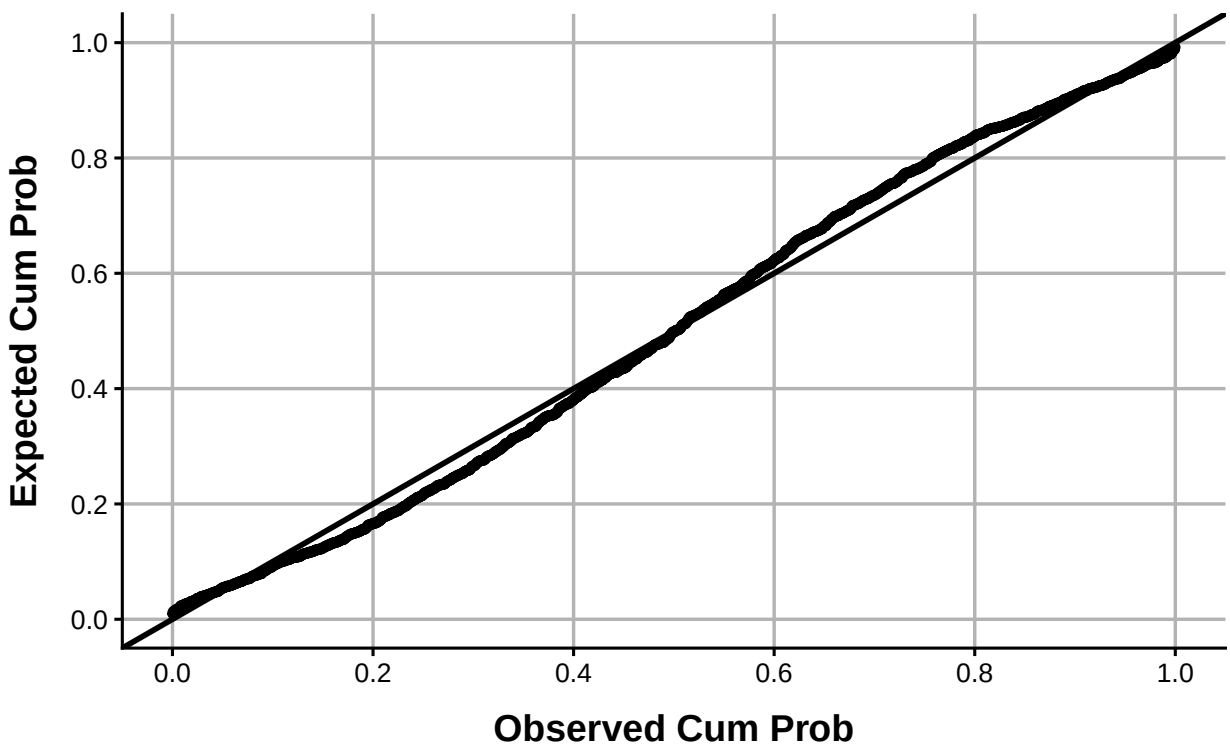
Residuals for LN_PRICE (Inliers) Model

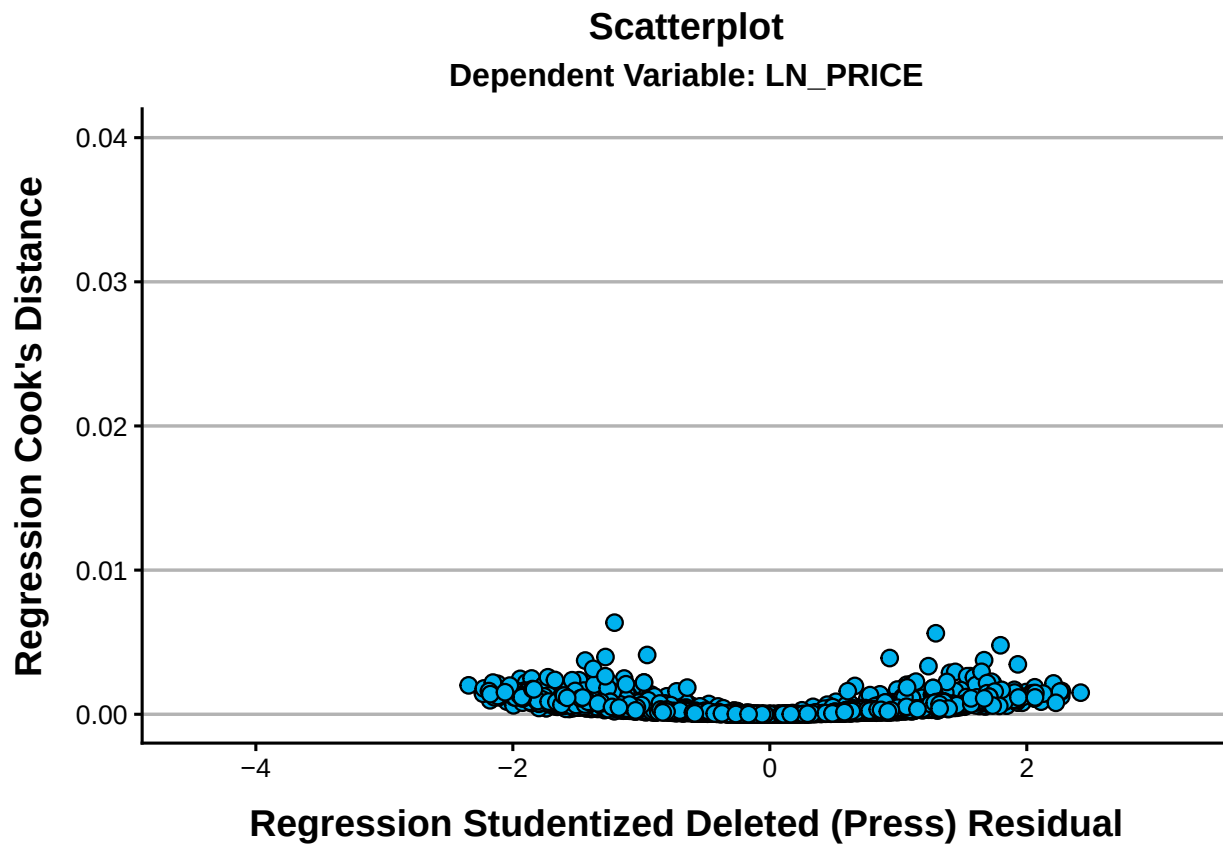
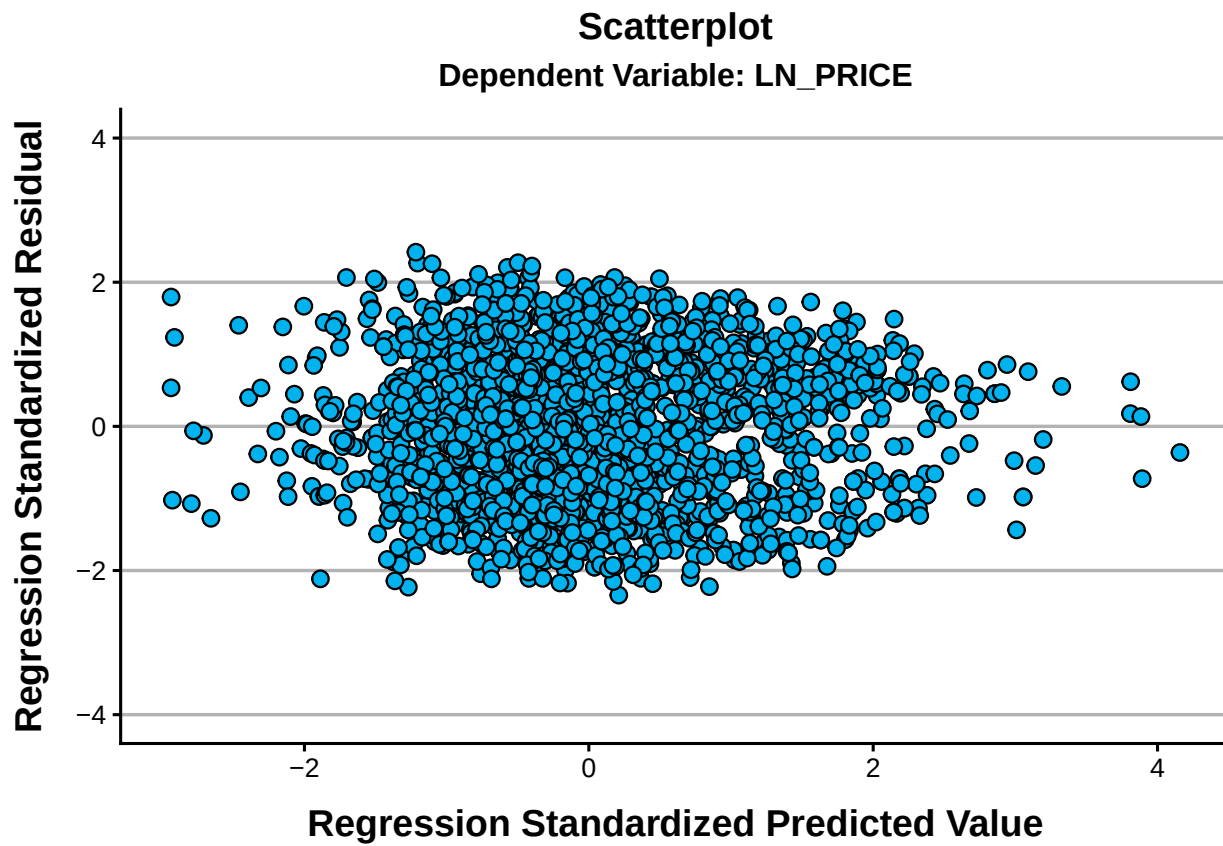


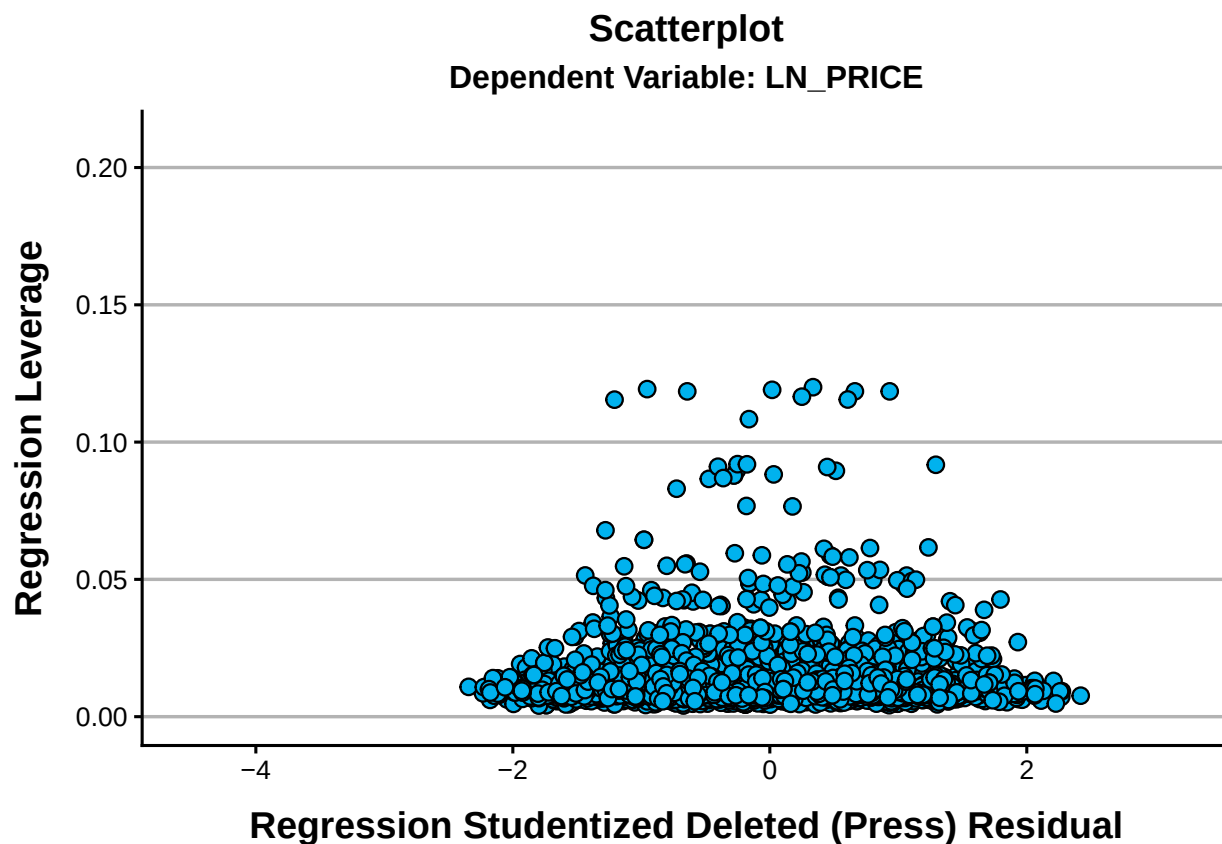
Histogram
Dependent Variable: LN_PRICE



Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE







Model IV

Table 10: District 4 LN_TASP Model I assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2264	1.086294	1.214818	0.9871254	1.230662	41.1898	-0.7317957

Table 11: District 4 LN_TASP Model II assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2264	1.089394	1.213866	0.9872134	1.229588	40.89744	-0.7279736

Table 12: District 4 LN_TASP Model I fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2130	2101	29	0.4143	0.4506	0.4433	0.4115	0.347	61.5389	0	2321.906	2491.822

Table 13: District 4 LN_TASP Model I coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.9676	321.0639	0.0342	0.0000	10.9007	11.0346
ln_landratio	0.1104	1.9974	0.0553	0.0459	0.0020	0.2188

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ln_base_bldgsize_ratio	0.5467	9.0904	0.0601	0.0000	0.4287	0.6646
single_family1story	-0.0765	-3.1379	0.0244	0.0017	-0.1244	-0.0287
half_duplex1story	-0.1687	-2.4815	0.0680	0.0132	-0.3021	-0.0354
two_plus_baths	0.2437	2.7669	0.0881	0.0057	0.0710	0.4165
good	0.2895	3.3802	0.0856	0.0007	0.1215	0.4574
fair	-0.1639	-7.2154	0.0227	0.0000	-0.2085	-0.1194
poor	-0.2474	-3.7883	0.0653	0.0002	-0.3754	-0.1193
very_poor_unsound	-0.3299	-2.7318	0.1208	0.0064	-0.5667	-0.0931
qbelow_avg	-0.0607	-2.3122	0.0263	0.0209	-0.1123	-0.0092
yb_1946_to_1960	0.0492	2.1371	0.0230	0.0327	0.0041	0.0943
ln_garagespaces	0.0659	2.4603	0.0268	0.0140	0.0134	0.1185
fireplace	0.0949	4.4833	0.0212	0.0000	0.0534	0.1365
wall	-0.2827	-2.0018	0.1412	0.0454	-0.5596	-0.0057
central_air	0.0887	3.6050	0.0246	0.0003	0.0405	0.1370
ecf_4R402	-0.4780	-5.7933	0.0825	0.0000	-0.6398	-0.3162
ecf_4R404	0.1865	3.7429	0.0498	0.0002	0.0888	0.2841
ecf_4R405	0.2493	6.9796	0.0357	0.0000	0.1793	0.3194
ecf_4R406	0.1736	5.0466	0.0344	0.0000	0.1061	0.2410
ecf_4R408	0.0969	2.5227	0.0384	0.0117	0.0216	0.1722
ecf_4R409	0.1563	2.6104	0.0599	0.0091	0.0389	0.2737
ecf_4R410	0.4532	13.7927	0.0329	0.0000	0.3887	0.5176
ecf_4R413	0.2859	7.4952	0.0381	0.0000	0.2111	0.3607
ecf_4R414	0.6958	12.8360	0.0542	0.0000	0.5895	0.8021
ecf_4R415	0.7637	12.2081	0.0626	0.0000	0.6410	0.8863
ecf_4R417	0.4538	12.0697	0.0376	0.0000	0.3801	0.5275
ecf_4R419	-0.2209	-3.4456	0.0641	0.0006	-0.3466	-0.0952
ecf_4R422	0.7113	13.2556	0.0537	0.0000	0.6061	0.8165

Table 14: District 4 LN_TASP Model II fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2130	2100	30	0.4137	0.4526	0.445	0.4107	0.346	59.8701	0	2316.139	2491.72

Table 15: District 4 LN_TASP Model II coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	9.0686	35.0449	0.2588	0.0000	8.5611	9.5761
ln_landratio	0.1161	2.1153	0.0549	0.0345	0.0085	0.2237
lnsbldsfint	0.2930	7.5487	0.0388	0.0000	0.2169	0.3691
single_family1story	-0.0740	-3.2309	0.0229	0.0013	-0.1189	-0.0291
two_plus_baths	0.3330	3.7570	0.0886	0.0002	0.1592	0.5069
powder_room	0.0672	2.1086	0.0319	0.0351	0.0047	0.1298
good	0.2922	3.4194	0.0855	0.0006	0.1246	0.4598
fair	-0.1640	-7.2249	0.0227	0.0000	-0.2085	-0.1195
poor	-0.2418	-3.7085	0.0652	0.0002	-0.3697	-0.1139
very_poor_unsound	-0.3443	-2.8568	0.1205	0.0043	-0.5806	-0.1079
qbelow_avg	-0.0746	-2.7297	0.0273	0.0064	-0.1283	-0.0210
yb_1946_to_1960	0.0489	2.1339	0.0229	0.0330	0.0040	0.0938
garage	0.0636	2.8784	0.0221	0.0040	0.0203	0.1069
fireplace	0.0894	4.1748	0.0214	0.0000	0.0474	0.1313
wall	-0.2728	-1.9353	0.1410	0.0531	-0.5492	0.0036
central_air	0.0866	3.5254	0.0246	0.0004	0.0384	0.1347
ecf_4R402	-0.4652	-5.6262	0.0827	0.0000	-0.6273	-0.3030
ecf_4R403	0.0908	1.7691	0.0513	0.0770	-0.0099	0.1914

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_4R404	0.2122	4.1374	0.0513	0.0000	0.1116	0.3128
ecf_4R405	0.2603	7.0461	0.0369	0.0000	0.1878	0.3327
ecf_4R406	0.1903	5.3328	0.0357	0.0000	0.1203	0.2602
ecf_4R408	0.1099	2.7930	0.0393	0.0053	0.0327	0.1870
ecf_4R409	0.1696	2.8000	0.0606	0.0052	0.0508	0.2884
ecf_4R410	0.4670	13.7522	0.0340	0.0000	0.4004	0.5336
ecf_4R413	0.2967	7.6498	0.0388	0.0000	0.2207	0.3728
ecf_4R414	0.7041	12.8515	0.0548	0.0000	0.5966	0.8115
ecf_4R415	0.7684	12.1874	0.0630	0.0000	0.6448	0.8920
ecf_4R417	0.4644	12.0581	0.0385	0.0000	0.3889	0.5400
ecf_4R419	-0.2236	-3.4617	0.0646	0.0005	-0.3502	-0.0969
ecf_4R422	0.7286	13.4335	0.0542	0.0000	0.6222	0.8350

